
Test Plan

for

Mentor Caring System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
B09	2026.05.14	Initial Content	1.0

1. Objectives

1.1 Modules under test

This test plan covers three main use cases of the Mentor Caring System: Search Student's Info, Edit Interview Record, and Search Mentor Info.

1.1.1 Search Student's Info

This module allows Mentors and MCP Coordinators to search student information by student ID. The test focuses on successful search, invalid or empty input, non-existing student ID, incorrect student ID format, and role-based access control. Faculty Consultant's Search Student's Info function is not included in this test plan according to the teacher's clarification.

1.1.2 Edit Interview Record

This module allows Mentors to add, modify, delete, save, or cancel interview records after a student record has been found. The test focuses on valid editing operations, invalid input, required fields, cancel operation, unauthorized access, and system behavior under load.

1.1.3 Search Mentor Info

This module allows Faculty Consultants and MCP Coordinators to search mentor information by mentor name, email, or group ID, and view related group members and student records. The test focuses on successful search, invalid or empty input, no matching result, group selection, and role-based access control.

1.2 References

The following documents and materials are used as references for this test plan.

1. Lab 11 Tasks PPT
2. Test Plan Template
3. Software Requirements Specification, Version 1.4
4. UI Diagram
5. Project Long Description, Version 1.2

2. Testing levels and methods

2.1 Testing levels

The main testing level used in this test plan is system testing. System testing is selected because the tests are designed based on complete use cases instead of individual classes or methods. The selected use cases are Search Student's Info, Edit Interview Record, and Search Mentor Info. Each use case is tested from the user's point of view and may involve the user interface, backend logic, database access, and role-based authorization.

Integration between the user interface, system logic, and database may also be involved during test execution, but it is not treated as a separate testing level in this plan. Unit testing is not included because this plan does not focus on individual functions, classes, or methods. Regression testing may be performed after bug fixes, but it is not the main focus of this initial test plan.

2.2 Testing methods

The testing methods used in this test plan include black-box testing, equivalence class testing, boundary value testing, state-transition-based testing, GUI testing, and load testing.

Black-box testing is the main method. Test cases are designed based on user inputs, user actions, and expected system outputs, without checking the internal source code.

Equivalence class testing is used to cover valid and invalid input classes, such as valid IDs, non-existing records, invalid formats, empty input, and invalid interview record data.

Boundary value testing is used where input limits are defined, such as empty values, minimum required input, maximum allowed input, and input exceeding the limit.

State-transition-based testing is used to cover main user flows and alternative flows, including Search, Edit, Save, Cancel, Back, Show Members, and selecting student records.

GUI testing is used for Search Student's Info to verify that the search process can be completed correctly through the system interface.

Load testing is used for Edit Interview Record to observe system response time, errors, and stability when multiple users perform the process at the same time.

Search Mentor Info is tested through functional black-box testing, including valid search, invalid input, no-result cases, group/member display, and navigation.

3. Test cases

This section describes the test cases for the selected modules of the Mentor Caring System.

3.1 Test cases for Search Student's Info

Actors tested: Mentor, MCP Coordinator

Tool involved: TestComplete

Not included: Faculty Consultant's Search Student's Info

Test ID	Test case description	Expected result
SSI-01	Mentor searches for an existing student using a valid student ID within the mentor's own group.	The system displays the student's basic information and interview records in read-only mode.
SSI-02	Mentor searches for an existing student using a valid student ID outside the mentor's own group.	The system denies access and displays an authorization warning message.
SSI-03	MCP Coordinator searches for an existing student within the coordinator's authorized department or scope.	The system displays the student's basic information and interview records in read-only mode.
SSI-04	MCP Coordinator searches for an existing student outside the coordinator's authorized department or scope.	The system denies access and displays an authorization warning message.
SSI-05	User searches using a non-existing student ID.	The system displays a clear error message indicating that no matching student record is found.
SSI-06	User leaves the student ID field empty and clicks Search.	The system displays a warning message and remains on the Search Student Info page.
SSI-07	User enters an invalid student ID format and clicks Search.	The system rejects the input and displays a warning message.
SSI-08	User clicks Back or Home from the Search Student Info page.	The system returns to the previous page or the corresponding home page.
SSI-09	TestComplete records and replays a valid Search Student's Info operation.	The recorded GUI test runs successfully and the expected student information is displayed.
SSI-10	User enters a student ID whose format is incorrect and clicks Search.	The system rejects the incorrect ID format, displays a clear format warning message, and remains on the Search Student Info page.

3.2 Test cases for Edit Interview Record

Actor tested: Mentor

Tool involved: LoadRunner

Precondition: The mentor has logged in and has already searched for a student in the mentor's own group.

Test ID	Test case description	Expected result
EIR-01	Mentor opens an existing student record and clicks Edit.	The system enables editing controls for the interview records and displays Save and Cancel controls.
EIR-02	Mentor adds a new interview record with valid interview date and time, problem statement, interview summary, and follow-up actions, then clicks Save.	The system saves the new interview record and refreshes the Student Info page with the latest content.
EIR-03	Mentor modifies an existing interview record with valid information and clicks Save.	The system saves the modified interview record and displays the updated content.
EIR-04	Mentor deletes an existing interview record and clicks Save.	The system removes the selected interview record and displays the updated record list.
EIR-05	Mentor edits an interview record but clicks Cancel instead of Save.	The system discards all unsaved changes and restores the previous content.
EIR-06	Mentor leaves a required field empty and clicks Save.	The system displays a validation warning and does not save the incomplete record.
EIR-07	Mentor enters an invalid interview date or time format and clicks Save.	The system rejects the invalid date or time and displays a warning message.
EIR-08	Mentor enters text exceeding the allowed length in problem statement, interview summary, or follow-up actions.	The system rejects the input or displays a length limit warning.
EIR-09	A non-mentor user opens a student record and attempts to edit the interview record.	The system keeps the record in read-only mode and denies the editing action.
EIR-10	Mentor saves an edited interview record and then searches the same student again.	The updated interview record is displayed correctly.
EIR-11	Multiple virtual users run the Edit Interview Record process using LoadRunner.	The system remains available, no critical error occurs, and system response time is observed.
EIR-12	LoadRunner test finishes and the result is opened in Analysis.	The test result summary, performance graph, or report is available for review.

3.3 Test cases for Search Mentor Info

Actors tested: Faculty Consultant, MCP Coordinator

Tool involved: No specific testing tool required by Lab 11 for this use case.

Test ID	Test case description	Expected result
SMI-01	Faculty Consultant searches for mentor information using a valid mentor name.	The system displays matching mentor information, including mentor name and associated group ID.
SMI-02	Faculty Consultant searches for mentor information using a valid mentor email.	The system displays matching mentor information.
SMI-03	Faculty Consultant searches for mentor information using a valid group ID, if group ID search is implemented.	The system displays the mentor associated with the group ID.
SMI-04	MCP Coordinator searches for mentor information within the coordinator's authorized department or scope.	The system displays matching mentor information and related group information.
SMI-05	MCP Coordinator searches for mentor information outside the coordinator's authorized department or scope.	The system denies access or does not display unauthorized information.
SMI-06	User searches using invalid or non-existing mentor information.	The system displays an error message and remains on the Mentor Search page.
SMI-07	User leaves all search fields empty and clicks Search.	The system displays a warning message and remains on the Mentor Search page.
SMI-08	User searches successfully, selects a group ID, and clicks Show Members.	The system displays the mentor information and the member list of the selected group.
SMI-09	User clicks Show Members without selecting a group ID.	The system displays a warning message and remains on the Mentor Result page.
SMI-10	User selects a student name from the member list.	The system displays the selected student's detailed record.
SMI-11	User clicks Search Again or Back from the result page, member list page, or student record page.	The system returns to the previous page according to the navigation path.

A. Appendix

This appendix provides screenshot evidence for the TestComplete GUI testing activity. The screenshots support the Search Student's Info use case, including recorded keyword test steps, property checkpoints, and the final test log.

A.1 Recorded Keyword Test Steps

Figure 1 shows the recorded Keyword Test steps in TestComplete. The recorded steps include browser navigation, page operations, student ID input, and search execution.

A. The recorded Keyword Test steps

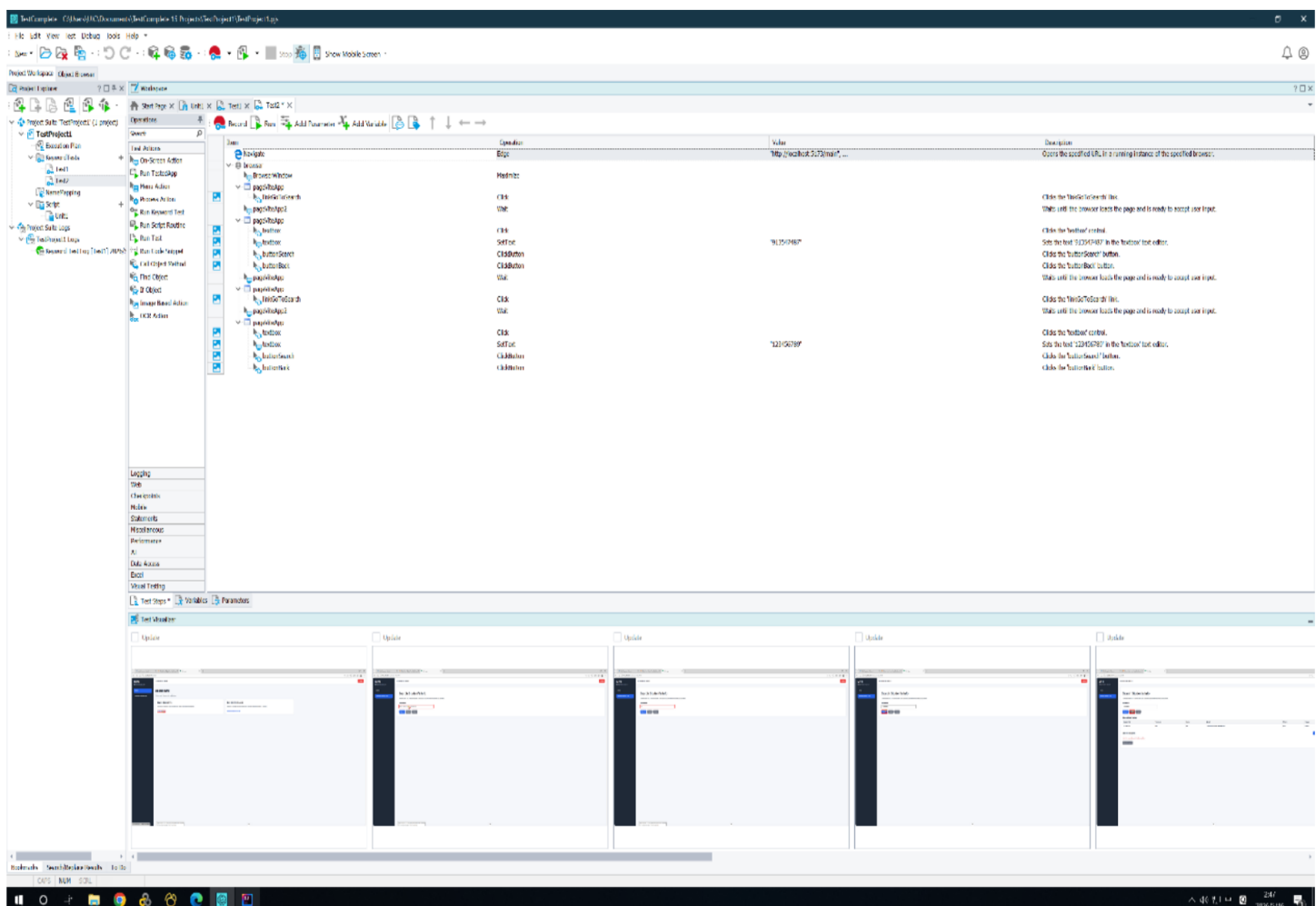


Figure 1: Recorded Keyword Test Steps for Search Student's Info

A.2 Checkpoint for Valid Student Information

Figure 2 shows the property checkpoint setting for the valid student ID search result. The checkpoint is used to verify that the expected student information is displayed after a valid student ID is searched.

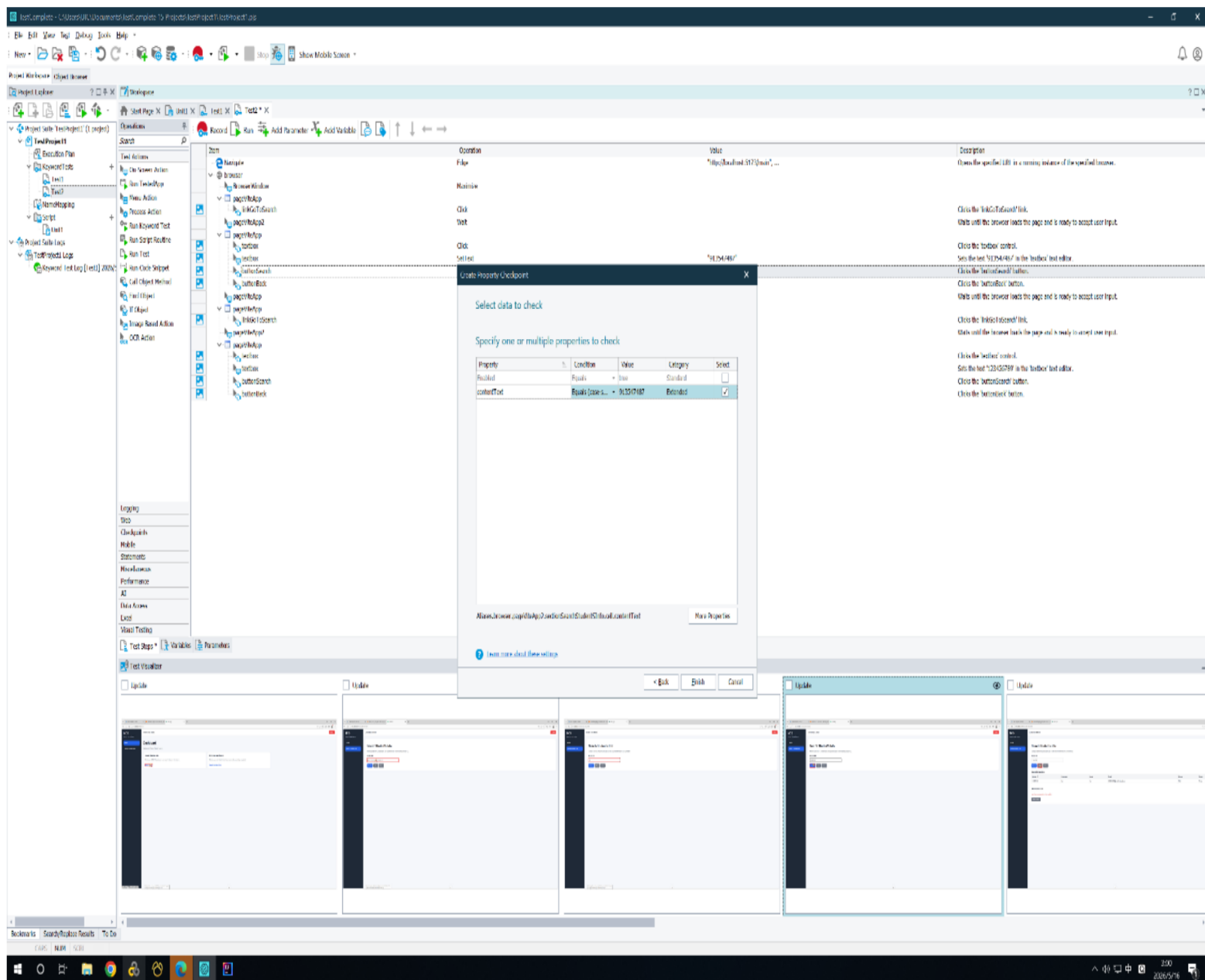


Figure 2: Property Checkpoint for Valid Student Information Display

A.3 Checkpoint for Warning Message

Figure 3 shows the property checkpoint setting for an invalid or empty student ID input. The checkpoint is used to verify that the system displays an appropriate warning message when the search input is invalid or empty.

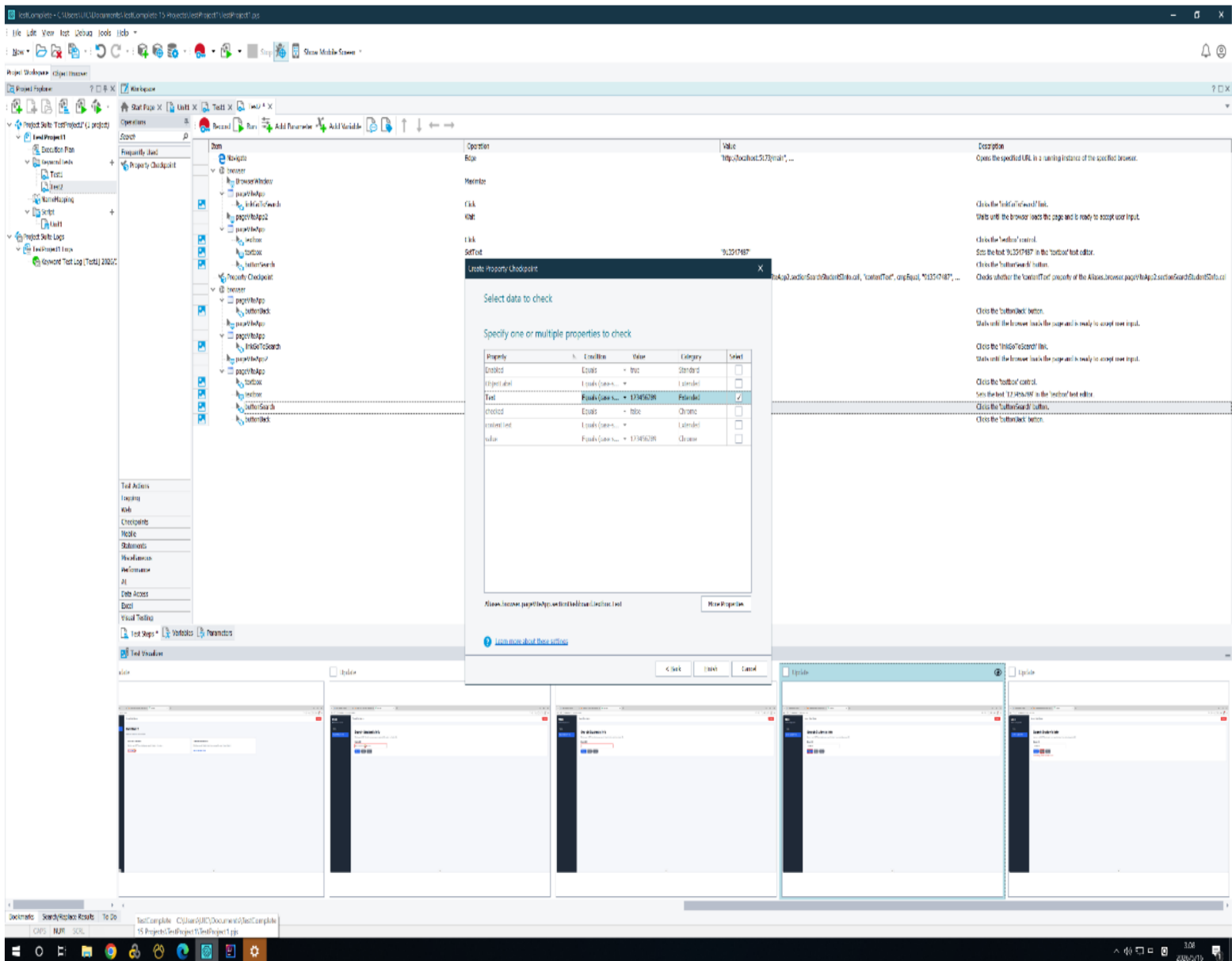


Figure 3: Property Checkpoint for Invalid or Empty Student ID Warning Message

A.4 Keyword Test Log Result

Figure 4 shows the TestComplete test log after running the keyword test. The log shows the execution result and confirms that the property checkpoint steps passed during the GUI test.

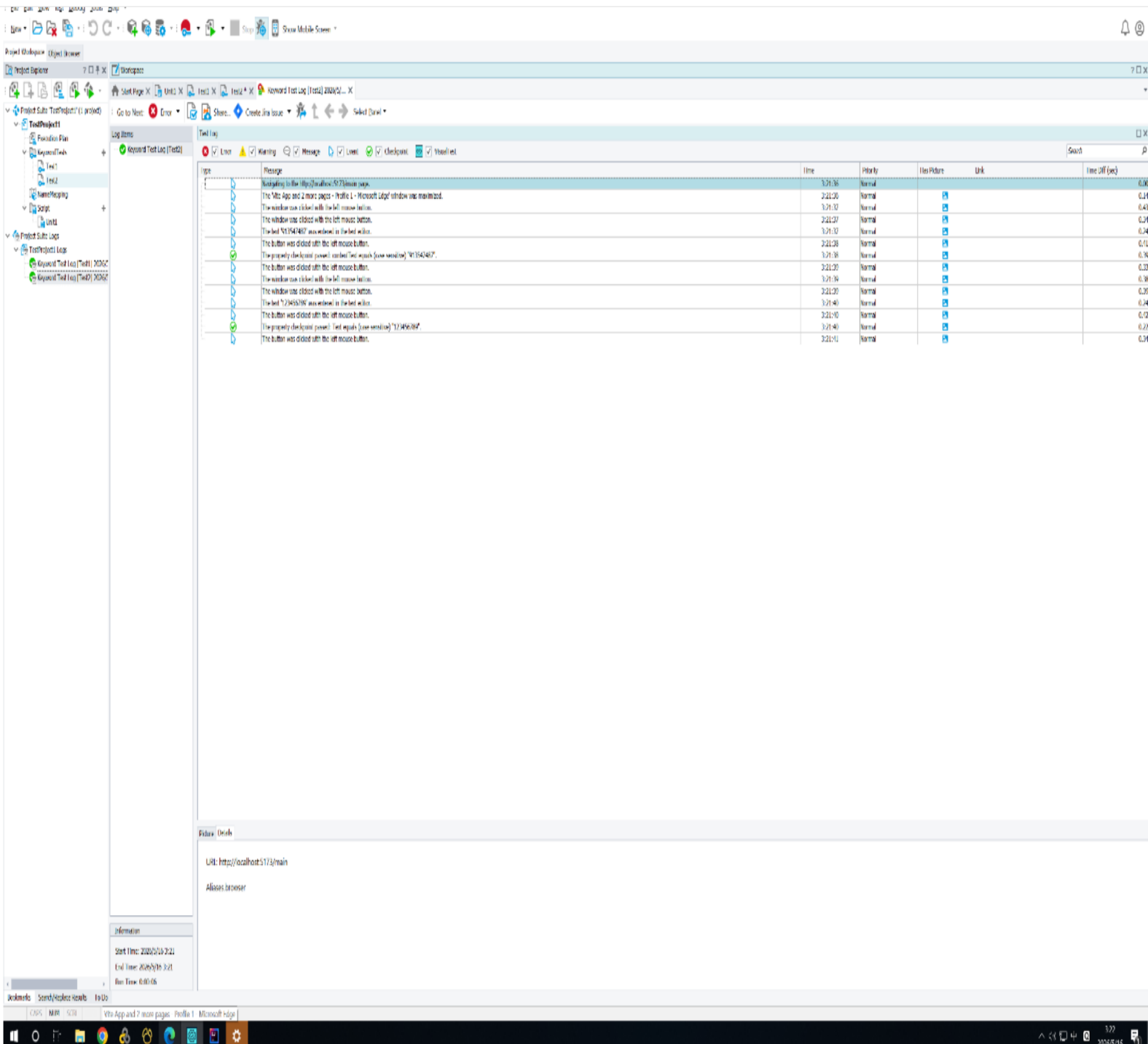


Figure 4: Keyword Test Log Showing Passed Checkpoint Results